Edge AI Benchmarking Framework for Performance Evaluation Across CPU, GPU, and FPGA Platforms

Parth Sharma, Department of Computer and Electrical Engineering
Dr. Biresh Kumar Joardar, Department of Computer and Electrical Engineering

Introduction

- AI applications are increasingly deployed on resource-constrained edge devices such as smartphones and embedded systems.
- Efficient on-device inference requires evaluating performance across different execution platforms.
- This project develops an Android benchmarking application to measure latency, memory usage, temperature, and battery consumption for multiple AI models.
- A Linear Regression model was also implemented on a Xilinx ZedBoard FPGA to investigate hardware acceleration for edge AI workloads.

Methods

Task 1: Android Benchmarking Application

- Developed an Android application for benchmarking AI inference on mobile devices.
- Implemented five AI models
  - Linear Regression
  - K-Nearest Neighbors (KNN)
  - Decision Tree
  - Graph Neural Network (GCN/GNN)
  - Custom 4-layer CNN (VGG-inspired)

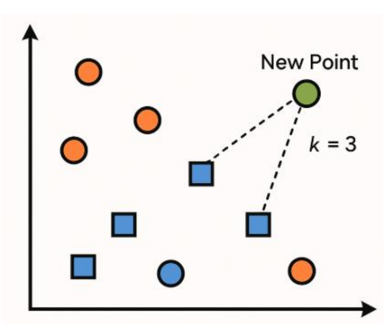
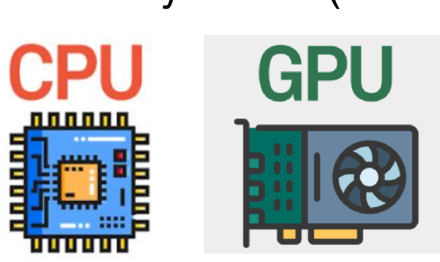


Figure 1: KNN



Supported Hardware:

1) Smartphone CPU
2) Smartphone GPU

- Records
  - Inference latency
  - Memory Usage
  - Device temperature
  - Battery consumption
  - Per-sample benchmark logs

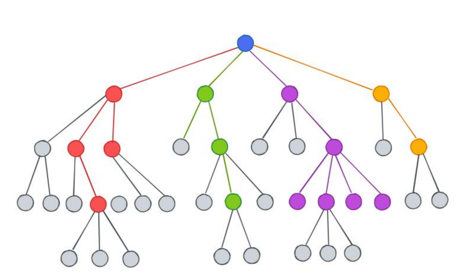


Figure 2: Decision tree

Task 2: FPGA Implementation

- Implemented the Linear Regression model on a Xilinx ZedBoard FPGA.
- Developed using:
  - Vitis HLS
  - Vivado
  - Vitis

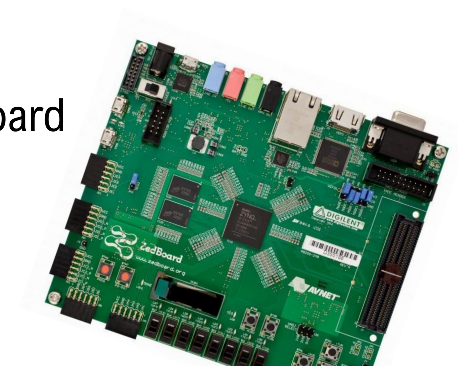


Figure 3: ZedBoard FPGA

For the FPGA implementation, the Linear Regression model was synthesized and deployed on a Xilinx ZedBoard to compare specialized hardware performance against conventional mobile processors.

Results and Analysis

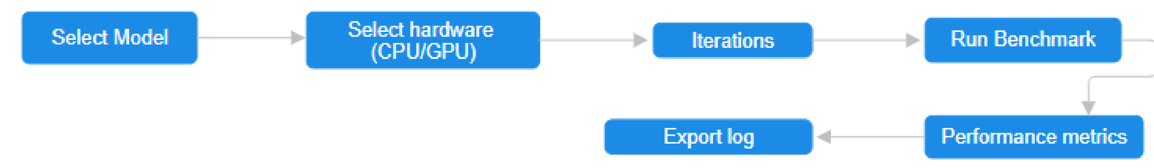


Figure 1: App Workflow

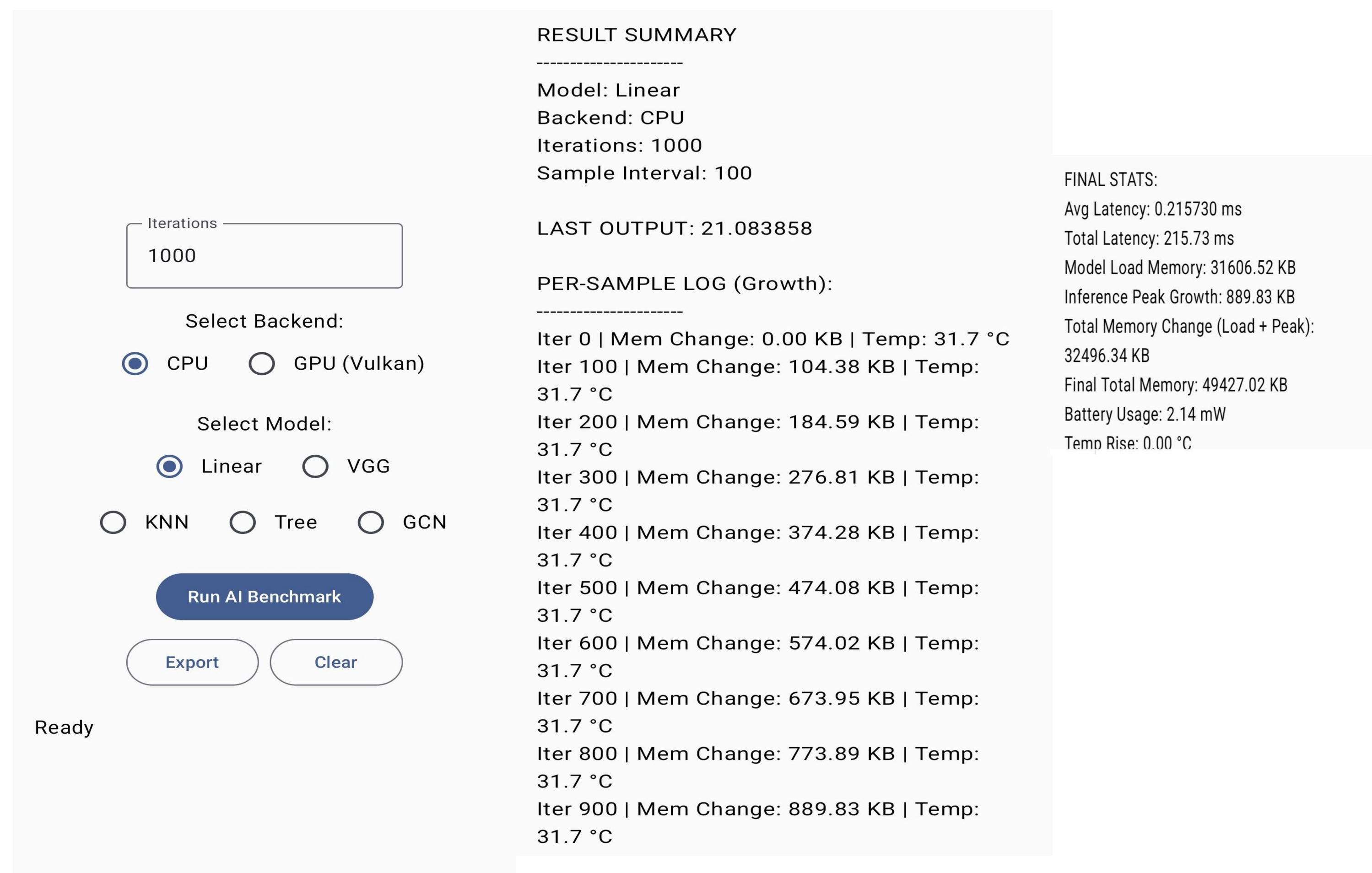
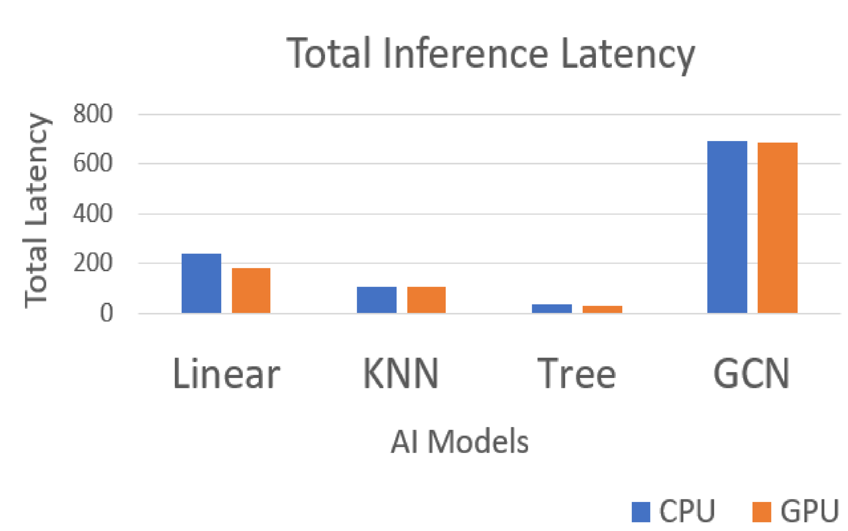
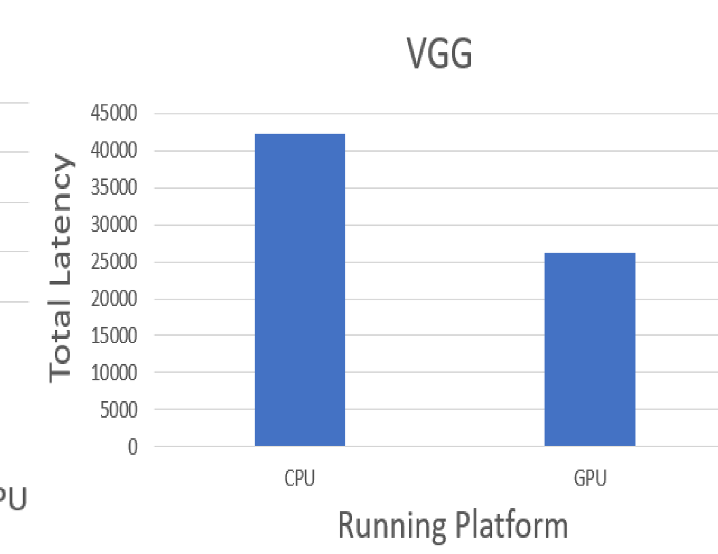
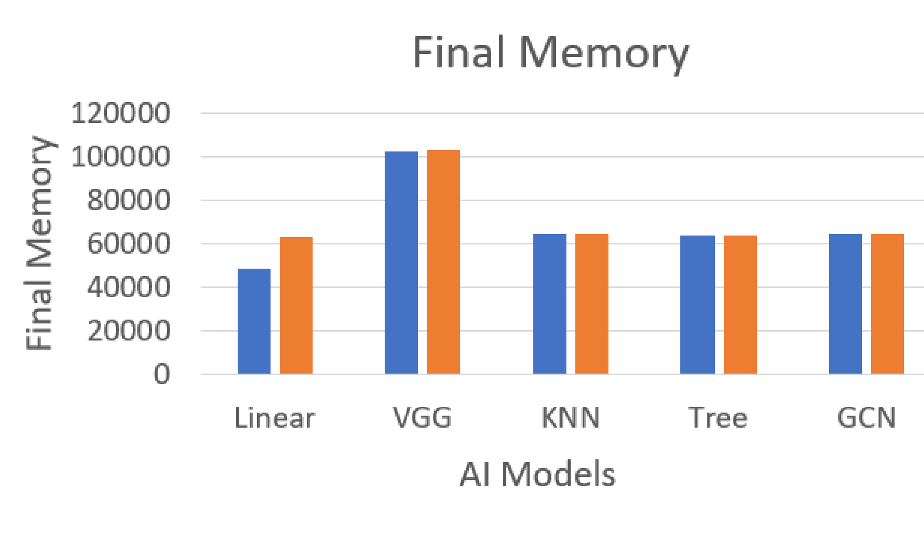


Figure 2: Phone App interface





- GPU execution consistently achieved lower inference latency than CPU across most evaluated AI models.
- More computationally intensive models (e.g., GCN and VGG) showed the largest performance improvements on the GPU.



- Memory utilization remained relatively stable across CPU and GPU executions for most models.
- Results indicate that latency improvements were achieved without a significant increase in memory consumption.

Conclusions

- Developed an Android benchmarking application for evaluating AI inference on mobile CPU and GPU platforms.
- Measured key performance metrics including latency, memory usage, temperature, and battery consumption.
- Implemented a Linear Regression accelerator on a Xilinx ZedBoard FPGA to demonstrate hardware-accelerated inference.
- The combined results provide insights into AI performance on both mobile processors and specialized hardware.
- The framework can be extended to benchmark additional AI models and edge computing platforms.

References

[1] K. Simonyan and A. Zisserman, "Very Deep Convolutional Networks for Large-Scale Image Recognition," *ICLR*, 2015. Available: https://arxiv.org/abs/1409.1556

[2] PyTorch Team, "PyTorch Mobile Documentation," PyTorch. Available: https://pytorch.org/mobile/home/

[3] AMD, "Vitis Unified Software Platform Documentation," AMD Xilinx. Available: https://docs.amd.com/r/en-US/ug1393-vitis-application-acceleration

Acknowledgements

Thank you so much to my faculty mentor, Dr. Biresh Kumar Joardar for their guidance, support and valuable feedback throughout this research project. I would also like to thank the SURF program and the Department of Computer end Electrical Engineering for providing the opportunity and resources to conduct this research.